

The impact of endovascular aneurysm repair on fluid dynamics in the abdominal aorta

G. Bruno 319812, A. Carta 324484, G. Galasso 319811, G. Ingrassia 331403, N. Migliore 332231, A. Villani 336800.

Abstract—This study aims to analyze the hemodynamic impact of endovascular aneurysm repair (EVAR) on fluid dynamics within the abdominal aorta using computational fluid dynamics (CFD). A patient-specific model of an abdominal aortic aneurysm was provided as an STL file and processed using CAD software. The aneurysm was virtually removed to simulate the post-operative geometry. Steady-state CFD simulations were performed in ANSYS Fluent by imposing two mean inlet velocity values. The simulations were carried out for both geometries (with and without aneurysm), resulting in four test cases. The analysis focused on velocity patterns, secondary flows, vorticity, local normalized helicity (LNH), and wall shear stress (WSS) compared for all models. Results indicate significant differences in flow behavior between aneurysmal and non-aneurysmal configurations, with vortex formation and lower WSS observed in the presence of an aneurysm, indices of disturbed flows. These findings demonstrate the importance of EVAR in restoring physiological flow conditions and potentially reducing risk factors for aneurysm progression or rupture.

Keywords—abdominal aortic aneurysm, CFD, endovascular repair, hemodynamics, wall shear stress

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The authors contributions are as follow:

G. Bruno – CFD simulation, extract parameters, writing

A. Carta – CAD modelling, writing, research of informations

G. Galasso – CFD simulation, extract parameters

G. Ingrassia – CAD modelling, writing, project amministration

N. Migliore – CFD simulation, extract parameters, research of informations

A. Villani – CFD simulation, writing

I. INTRODUCTION

Abdominal aortic aneurysms (AAAs) are pathological dilatations of the abdominal aorta, often associated with high morbidity and mortality in case of rupture. The occurrence of this pathology depends on multiple factors, including genetic predisposition; biochemical factors; inflammation and immune response; the presence of atherosclerotic plaques; and finally hemodynamic factors. They turn out to be asymptomatic until rupture, and the main risks associated with this pathology turn out to be intraluminal thrombus formation; dissection and rupture of the aorta. One of the main procedures for the treatment of this pathology is EVAR. This is a minimally invasive procedure that aims to isolate the aneurysm from the bloodstream, thus reducing the risk of rupture. Understanding the hemodynamic changes induced by EVAR is essential to evaluate its effectiveness and potential complications. CFD has emerged as a valuable tool to simulate and analyze blood flow in complex vascular geometries. This project aims to study the hemodynamic impact of EVAR by comparing pre- and post-operative models under different flow conditions.

II. METHODS

The work is divided into four different phases. The first involves CAD modeling, on SOLIDWORKS, of an STL file of the provided AAA. Then the surfaces are imported to ANSYS Fluent to create the mesh. The third step involves performing CFD simulation of the finite element models created. Finally, post processing is performed on PARAVIEW to extract the hemodynamic parameters of interest and perform a comparison between the models.

A. Geometry and Mesh Generation

A patient-specific STL model of an AAA is provided. Using SOLIDWORKS, to simulate the action of the EVAR, the aneurysmal bulge is manually removed. The removal is done by performing a bulge cut, maintaining the two portions of the physiological vessel. Since EVAR has the effect of diverting blood flow, blocking blood access into the aneurysm, it is chosen to recreate a physiological geometry, similar to the healthy vessel condition. To do this, a loft is created to join the two surfaces obtained by the previous cut, and in order not to

create an extremely simple surface, as a straight cylinder, a guideline is created to give a physiological curvature to the reconstructed component. The result obtained is a smooth vessel with a diameter of around 2.3mm.

The entire geometry is imported in ANSYS Fluent and scaled to obtain the real dimension of the vessel. The scale factor used for this model is 10, because when importing the value into Fluent, geometry is one order of magnitude smaller than the real value of the physiological section. Polyhedral elements are used for discretization, allowing for a robust and sufficiently accurate solution of the vessel. The number of polyhedral cells is around 430 thousand for the model with aneurysm. While the number for the model without aneurysm is around 340 thousand.

TABLE I

Local Scoped Sizing	Value with aneurysm (mm)	Value without aneurysm (mm)
Minimum	0.3	0.3
Maximum	1.1	1
Growth rate	1.2	1.2
Normal angle	25	25

The values in Table I refer to the parameters set for the creation of the mesh, based on bibliographic research [1].

B. Boundary conditions and Simulation Setup

Created the two meshes, a steady-state CFD simulations are conducted in ANSYS Fluent. The blood is modeled as a Newtonian fluid with constant density ($\rho=1060 \text{ kg/m}^3$) and viscosity ($\eta = 3.5 \cdot 10^{-3} \text{ Pa} \cdot \text{s}$). At the inlet, two velocity conditions are imposed:

TABLE II

Case	Velocity value [m/s]
V1	0.092
V2	0.15

Velocity condition is set directly and not the mass flow rate to ensure a stronger imposition by making sure that the velocity will always be the same along the vessel. The outlet pressure is set equal to 0 Pa and for the vessel walls are set no slip condition and rigid behavior. Simulations were run until residuals converged to acceptable levels, ensuring stability and accuracy. Once the convergence is reached, the quantities of interest are extracted. The simulation solution method used for pressure-velocity coupling is SIMPLEC.

C. Analysis Parameters

Post-processing analysis focused on multiple hemodynamic quantities allowing us to compare the flows established within the two models. The quantities of interest displayed with Paraview are:

- LNH
- WSS
- Vorticity field
- Velocity profiles and streamlines
- Secondary flows (qualitative analysis)

LNH is a measure of the spiralization of the flow: it combines the direction of velocity and vorticity. The value is normalized between -1 and +1. LNH is used instead of helicity because it is adimensional, normalized and can therefore be compared between different regions of the flow, or between patients/studies; it is more useful for visualization and qualitative analysis; it prevents variations in intensity from masking the geometric information of the flow. It is defined as the helicity normalized by the modules of velocity and vorticity. And it can be understood as the cosine of the angle between the two vectors.

$$LNH(x, t) = \frac{v(x, t) \cdot \omega(x, t)}{|v(x, t)| |\omega(x, t)|} \quad (1)$$

In addition, LNH rather than helicity was chosen to be analyzed because we imposed a stationary flow.

WSS is the force per unit area exerted by the blood flow on the vessel wall, tangent to the surface. The WSS is expressed as the scalar product between viscosity and the velocity gradient at the wall.

$$WSS = \mu \left(\frac{\partial u_t}{\partial n} \right) \quad (2)$$

Vorticity field measures how much the flow rotates around a point. It is a vector field that describes the local rotation of the fluid.

$$\omega = \nabla \times v \quad (3)$$

Velocity profiles represent the distribution of blood velocity within the cross-section of a vessel. In laminar flow, blood flows faster in the center of the vessel and slower near the walls, due to friction. This profile can change in the presence of stenosis, aneurysms, or bifurcations. Streamlines are imaginary lines that represent the direction of flow at each point in the vessel. They indicate the path followed by a fluid particle over time. Under normal conditions, streamlines are regular and parallel; in the presence of obstacles or malformations, they can become turbulent or disorganized.

Secondary flows are deviations of blood flow from the main direction, which develops across the cross-section of the vessel. They often occur in complex geometries, such as curves, bifurcations, or stenoses, where centrifugal forces or the interaction between the wall and fluid cause vortices or spirals to transverse the main flow.

All these parameters are often visualized to show the complexity of the flow and identify critical areas where adverse phenomena (thrombosis, aneurysmal growth, rupture) may occur.

A. LNH

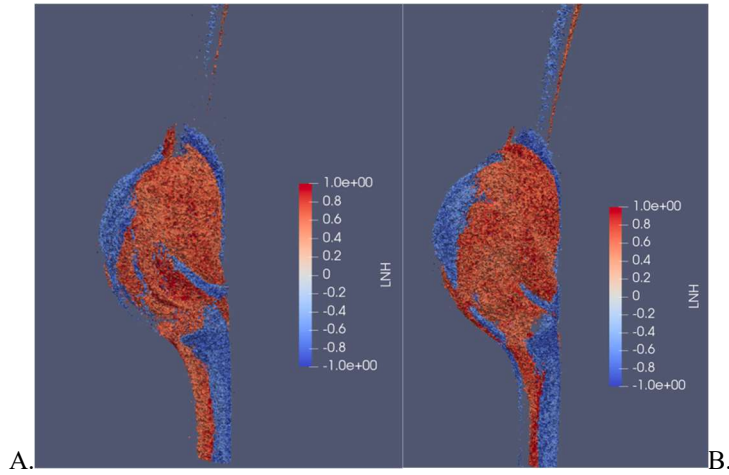


Figure 1: LNH with aneurysm (Fig. A: V1, Fig. B: V2)

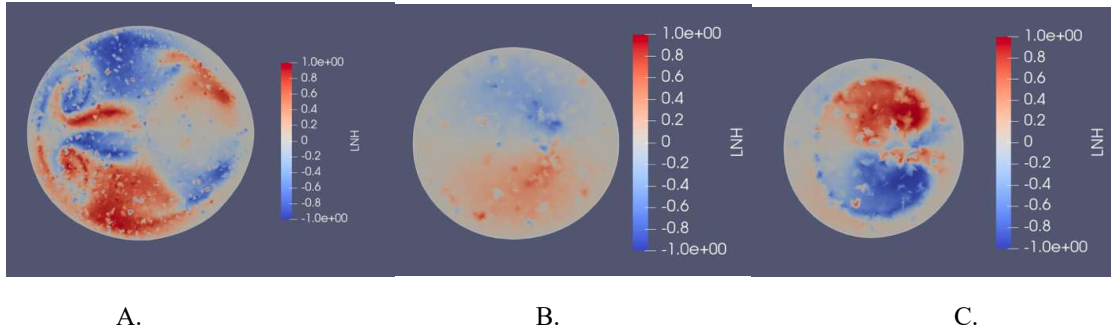


Figure 2: LNH section in the middle of the aneurysm (A) and sections proximal (B) and distal to the aneurysm (C) (case V1)

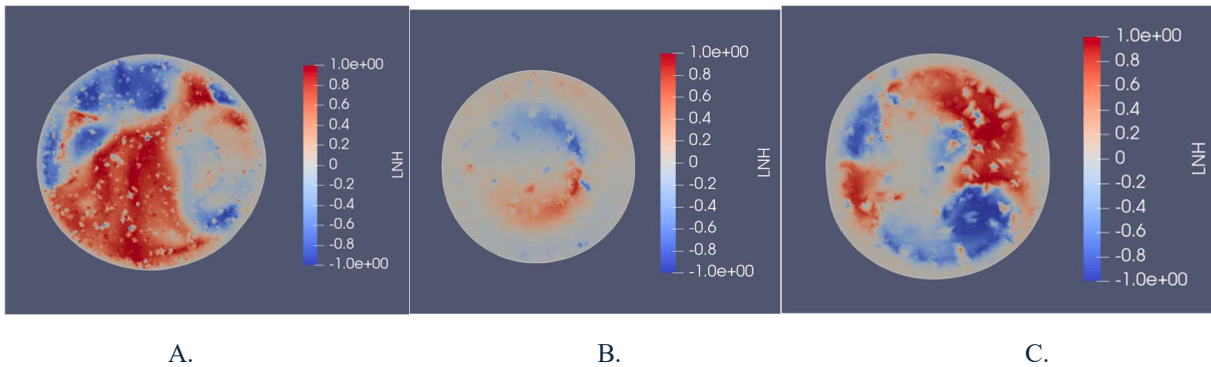


Figure 3: LNH section in the middle of the aneurysm (A) and sections proximal (B) and distal to the aneurysm (C) (case V2)

III. RESULTS

The results were obtained using PARAVIEW software. Following the CFD simulation performed on Fluent, hemodynamic quantities were extracted. Comparisons between the results were mainly focused on the relative part of vessel where the aneurysm is present and the same reconstructed part corresponding to an EVAR operation. The main extracted haemodynamic indices are shown below:

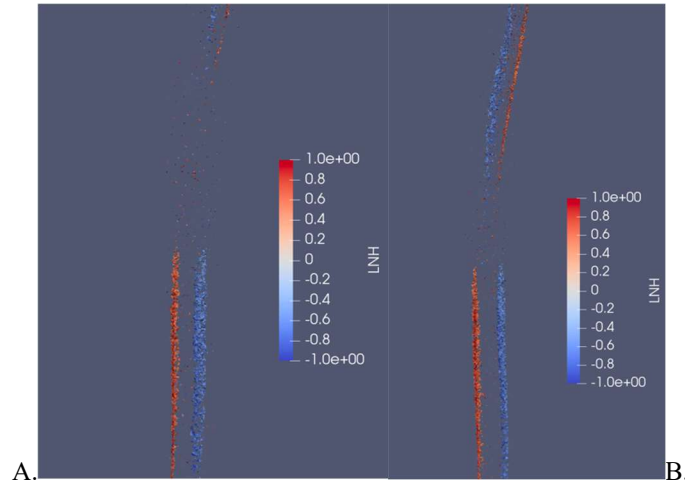


Figure 4: LNH in repaired model (Fig. A: V1, Fig. B: V2)

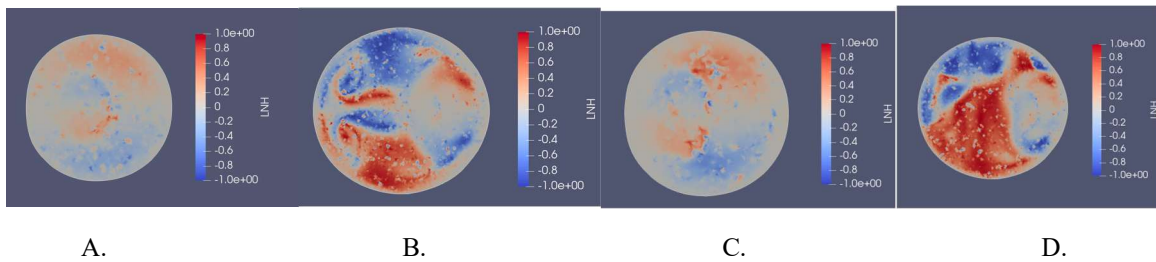


Figure 5: Cross sections comparison of the repaired part (A-C) and aneurysmal part (B-D) with the two velocities. A and B are associated with $v=0.092$ m/s; C and D are associated with $v=0.15$ m/s.

Regarding the LNH index, in the case of the aneurysm, the model with V1 exhibits a more organized helical flow respect to V2, where retrograde and antero-grade flow regions can be clearly distinguished. In contrast, the second model, characterized by a higher velocity V2, exhibits a more chaotic and turbulent LNH field, where the symmetry between the two flow regimes is lost (Fig.3). In particular, at V2 a supremacy of right-handed helical flow (vorticity in the same direction as the flow) is appreciated in figure 3A. In addition, it is observed that, in the sectioned part of the aneurysm (the largest

section is taken), more pronounced helical movements develop compared to the non-aneurysmal part (Fig.5). Moreover, in the sectioned portion of the repaired aortic segment, a flow pattern very similar to that observed upstream of the aneurysm is seen, characterized by lower LNH values indicative of a more physiological flow pattern (Fig.3B-2B and 5A-5C). In the aneurysmal region, LNH shows a mixture of both right-handed and left-handed helical flows, whereas downstream of the aneurysm, these flows tend to restore a physiological trend (Fig. 2C-3C)

B. WSS

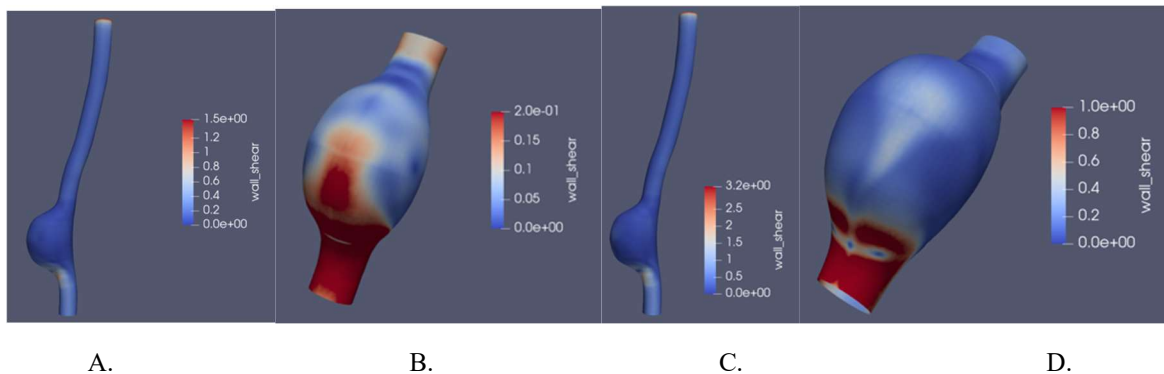


Figure 6: Sagittal plane of the WSS in model with aneurysm (A-C) and focus on the distribution of WSS on the aneurysm (B-D) (A-B refers to case V1; C-D refers to case V2)

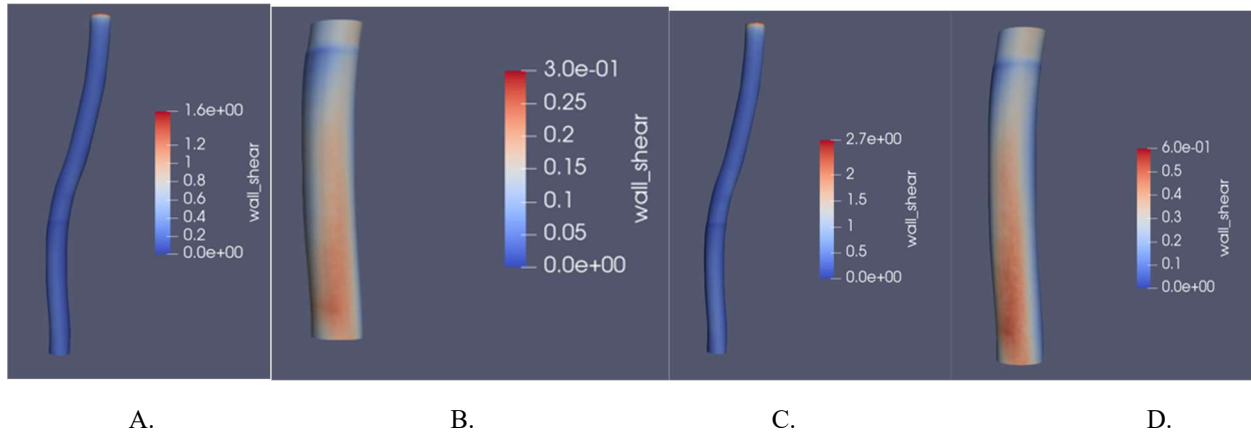


Figure 7: Sagittal plane of the WSS in repaired model (A-C) and focus on the distribution of WSS in repaired model (B-D) (A-B refers to case V1; C-D refers to case V2)

As far as the WSS is concerned, the maximum values presented by both models are similar at the same velocity (Fig. 6-7). The maximum value is observed at the inlet of the vessel due to the flat velocity profile. As the velocity increases, the WSS value tends to increase. Focusing on the section at the aneurysm (Fig. 6B-6D), it can be seen that the WSS generally takes on lower

values than the rest of the vessel, for both velocities, with the exception of the distal area of the aneurysm where we have high WSS values. In Fig. 7B-7D it can be seen that the EVAR process has restored evenly distributed physiological WSS values over the vessel wall.

C. Vorticity

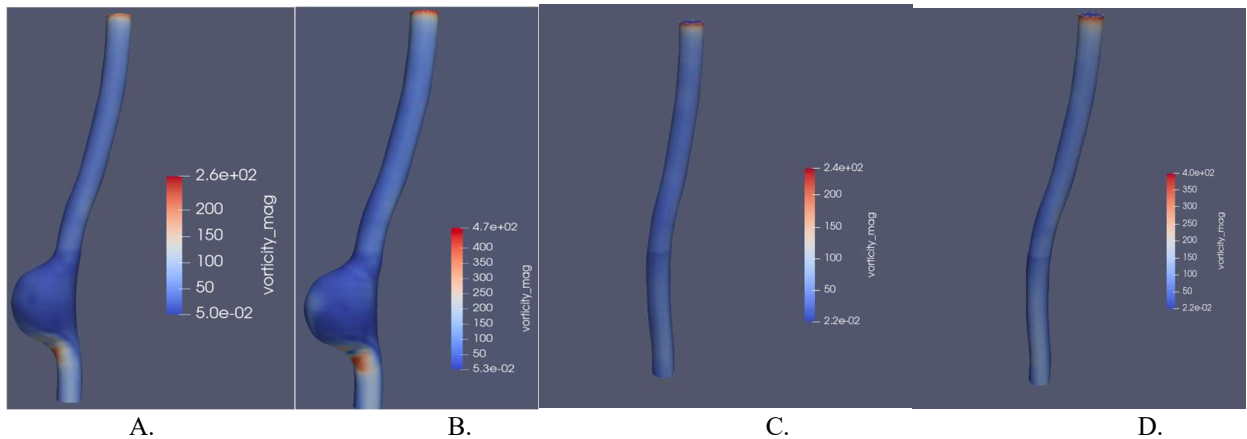


Figure 8: Sagittal plane of vorticity in model with aneurysm (Fig. A: V1, Fig. B: V2) and in repaired model (Fig. C: V1, Fig. D: V2)

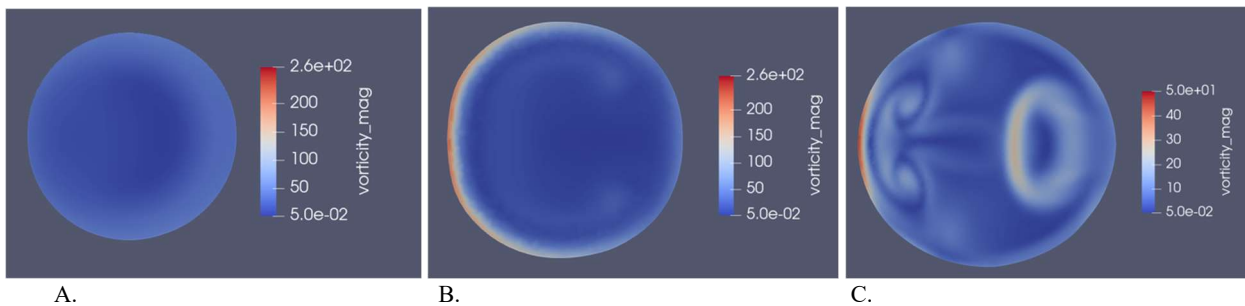


Figure 9: Vorticity in model with aneurysm, in area upstream (A), downstream (B) and cross section in the center (C) (case V1)

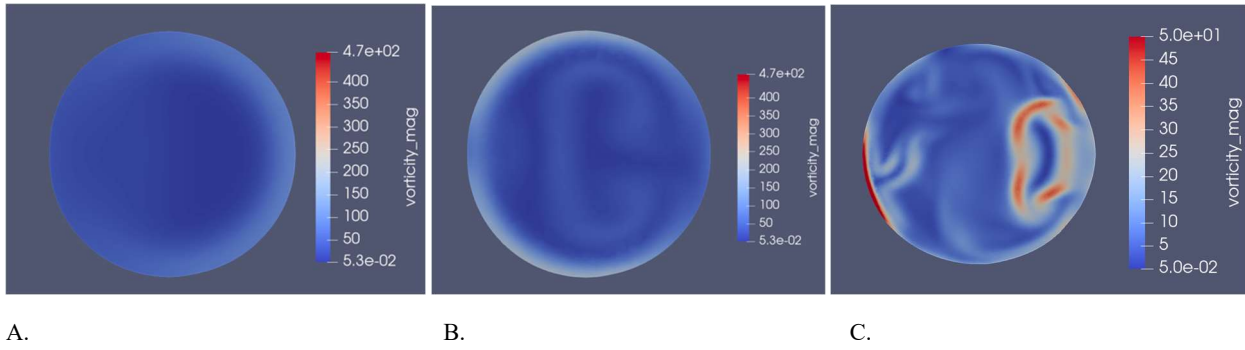


Figure 10. Vorticity in model with aneurysm, in area upstream (A), downstream (B) and cross section in the center (C) (case V2)

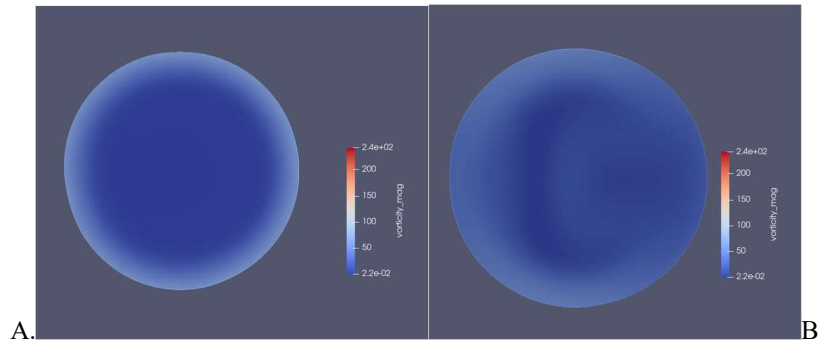


Figure 11. Vorticity in repaired model, in area upstream (A), downstream (B) (case V1)

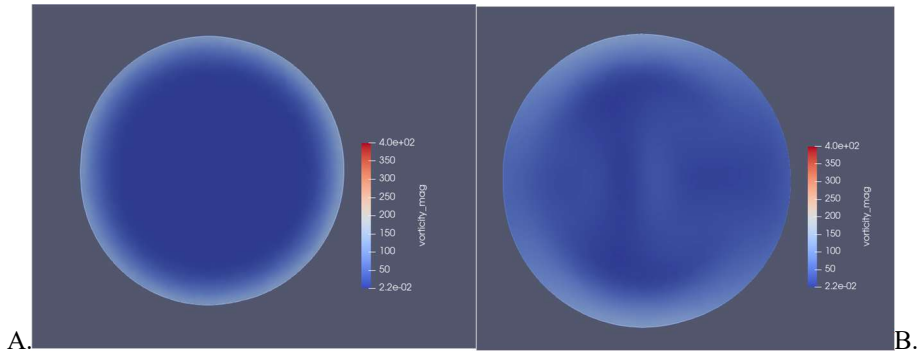


Figure 12. Vorticity in repaired model, in area upstream (A), downstream (B) (case V2)

Regarding vorticity, high values are observed especially near the vessel walls and just downstream of the dilated area (Fig.8A-8B). Greater vorticity is observed in the case of V2 compared to V1 (Fig.9-10) due to the higher kinetic energy of the fluid. Upstream of the aneurysm, vorticity is reduced, and the flow is more ordered (Fig.9A-10A). Downstream of the aneurysm, vorticity increases, with the presence of turbulent wake regions and flow instabilities (Figure 8A-8B-9B-10B). In the

presence of an aneurysm, flow separation zones originate. In the absence of an aneurysm, the smooth geometry prevents the formation of large recirculation zones. The flow follows more linear streamlines, and thus vorticity is reduced (Fig. 11-12). Overall, vorticity is higher, more disorganized, and concentrated near the aneurysmal bulge in the aneurysmal model, while it is lower and more evenly distributed in the non-aneurysmal case.

D. Velocity profiles

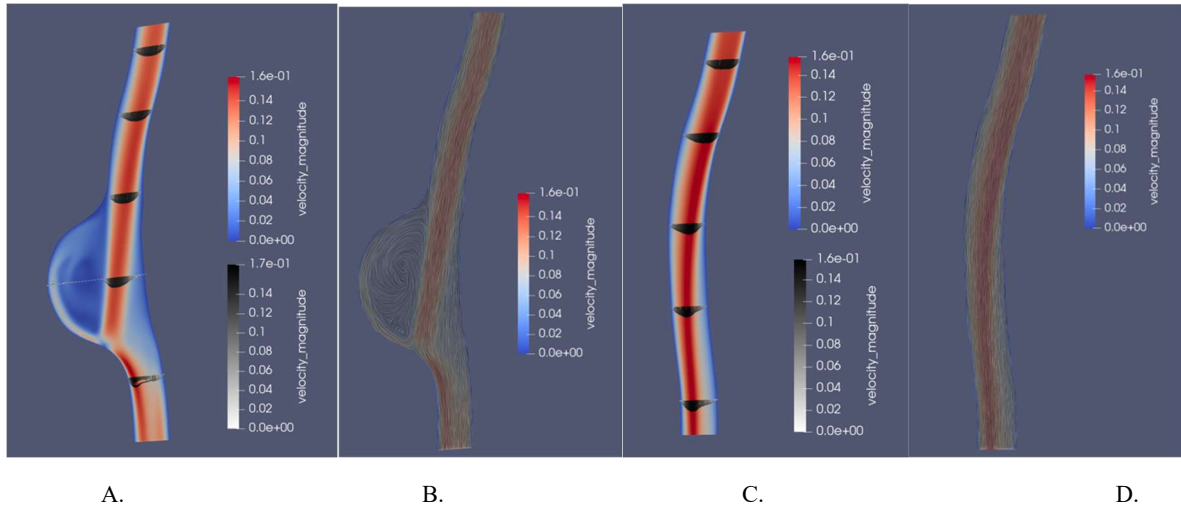


Figure 13. Sagittal section of the velocity magnitude with aneurysm (A-B) and without aneurysm (C-D) with the representation of the streamlines (case V1)

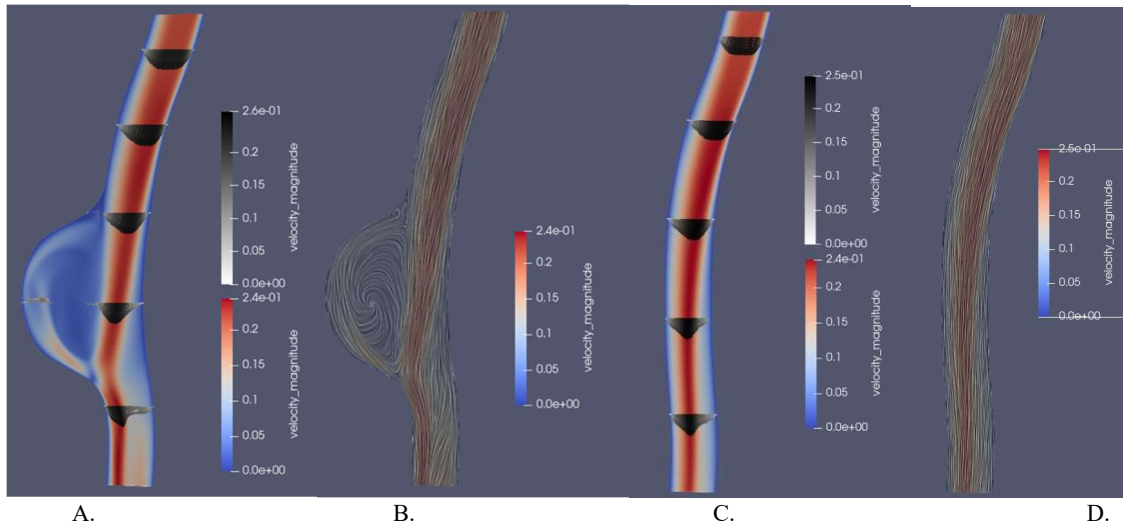


Figure 14. Sagittal section of the velocity magnitude with aneurysm (A-B) and without aneurysm (C-D) with the representation of the streamlines (case V2)

In the inlet section, a flat velocity profile is observed; as the flow progresses along the aortic segment, a parabolic profile develops, with maximum values at the center of the vessel and zero velocity at the wall, in agreement with the no-slip condition. The curvature of the vessel shifts the velocity peak away from the center, as shown in Fig.13A. Moreover in the aneurysmal part the velocity almost completely drops to zero in almost the entire section except in the central part which continues to follow the trend present above the aneurysm (Fig.13A and 14A), and then starts to redevelop downstream of the aneurysm. Furthermore, when the flow hits the final part of the aneurysmal belly it divides into two parts, one follows the lumen of the vessel, the other instead follows the aneurysmal wall causing an inversion of the flow and leading to areas of blood recirculation (Fig.13B and

14B). The velocity distribution in a vessel without aneurysm presents laminar flow with a parabolic profile, showing maximum velocity at the center of the lumen and zero values at the walls. The regular geometry allows the absence of turbulence or recirculation zones. It can be seen that in repaired duct, the velocity profile following a curve shifts the maximum peak toward the wall and then restores a perfectly parabolic profile (Fig.13C-D). Performing a comparison of the results obtained with two different velocities (Fig.13C and 14C), it can be seen that there are no significant differences between the two cases, both in the model with aneurysm and in the case without aneurysm. The only relevant factor is the velocity profile, which is more pronounced at higher velocities.

E. Secondary flows

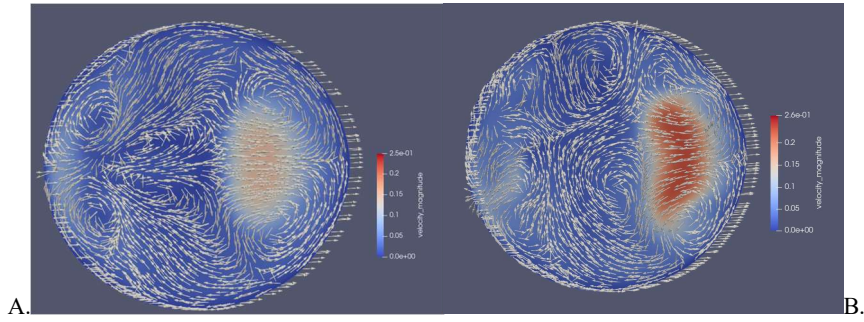


Figure 15. Section with aneurysm (fig. A: $V1$, fig. B: $V2$)

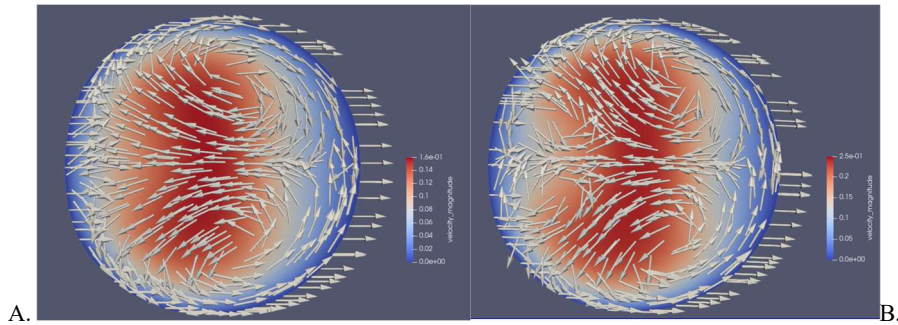


Figure 16. Section without aneurysm (fig. A: $V1$, fig. B: $V2$)

In the aneurysmal model, secondary flow patterns displayed complex vortex structures and noticeable recirculation zones (Fig.15). At the lower velocity ($v=0.092$ m/s), the vortices were more localized, while at the higher velocity ($v=0.15$ m/s), the flow disturbances and recirculation regions were more pronounced and widespread. In the post-EVAR model, secondary flows appeared more organized and symmetric (Fig.16). At both velocities, the reconstructed geometry resulted in reduced recirculation zones and more streamlined vortex structures compared to the aneurysmal model.

IV. DISCUSSION

The results obtained appear to be consistent with what has been verified in the literature, although they represent a simplified situation of reality.

LNH

As can be seen from Fig. 2A-3A the loss of helical coherence suggests increased local instabilities, typically associated with less physiological hemodynamic conditions. Within the aneurysm, the trend obtained is indicative of chaotic and disordered flow, predominantly at the “belly” of the aneurysm. According to a prior experimental study by Liu *et al.*[3] it is hypothesized that increased helical intensity may enhance expression of inflammatory biochemical factors on the degenerated AAA wall, increasing the risk of AAA rupture, which needs further validation in future experiments.

WSS

Lower WSS values are established within the bulge than the value obtained by repair, which promotes platelet aggregation and thrombus formation. These effects arise from the formation of swirling and disturbed flow patterns caused by changes in the vessel geometry. The presence of a low WSS, with values below the physiological range, is a possible indication of rupture of the aneurysmal wall, as it leads to its remodeling and thinning. The increase in WSS in the distal position is due to the velocity of flow going against the vessel wall. This occurs because there is a sharp transition from a smaller to a larger cross-sectional area, which leads to the formation of stagnation zones in the proximal position and a non-deflection of flow, which inevitably collides with the wall. Figures 7B-7D, representing the vessel following EVAR, show physiological WSS values that therefore reduce the possibility of damage to the endothelium and the corpuscular part of the blood. This certifies how EVAR enables restoration of proper hemodynamics in the vessel.

Vorticity

Regarding vorticity, high value corresponds to greater flow disturbance, which implies increased wall stress and may promote thrombosis or aneurysm progression. In the aneurysmatic models, the average vorticity is higher, and the flow appears more chaotic, due to the aneurysmal bulge being centered along the axis, with no marked anatomical asymmetries or lateral

bifurcations or curves. This explains why, in the presence of an AAA, the risk of thrombosis, inflammation, and rupture increases.

Velocity profiles

The flow, being laminar, as it enters the interior of the aneurysm, continues its direction as if it were in a cylindrical conduit, which is why the bulge-walled zones turn out to have non-high velocities and WSS. As can be seen from Fig. 13A-14A, the flow goes against the aneurysm wall, and this implies the formation of high WSS zones, as shown in Fig. 6B-6D, which turn out to be potentially dangerous as they lead to possible damage to the vessel wall. Fig. 13C-14C shows that the EVAR process re-obtained a parabolic velocity profile at the treated area, as there are no significant alterations in the vessel cross-section, but it is only affected by the slight curvature imposed by the reconstruction, that non significantly alters the profiles.

Secondary flows

The comparison between aneurysmal and post-EVAR models confirms that the aneurysm geometry promotes disturbed secondary flows, particularly at higher velocity ($v=0.15$ m/s), where chaotic vortices and central flow deviation were observed. These disturbed patterns are associated with low WSS and flow stagnation, conditions favoring thrombus formation and aneurysm progression [5]. Even at lower velocity ($v=0.092$ m/s), altered transverse flow persisted near the aneurysmal wall (Fig. 15). In contrast, the post-EVAR geometry restored organized and symmetric secondary vortices at both velocities (Fig. 16). The reconstructed curvature enabled a physiological double-vortex pattern, reducing recirculation and aligning with literature findings that associate flow normalization with reduced hemodynamic risk [9].

V. CONCLUSION

The study allowed us to verify the efficacy and hemodynamic benefits achievable with a revascularization process by comparing pre- and post-implantation parameters. The results obtained were found to be consistent with the literature, but it is important to specify that the results obtained should be interpreted with caution, as stringent simplifications were performed for the realization of the simulation. The main simplifications are steady-state flow, Newtonian fluid, and absent wall thickness. Further studies should focus on the effects of the EVAR process under pulsatile flow conditions; taking into consideration the wall thickness or modeling a device that actually simulates the presence of EVAR, which was absent in our study.

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